

TIDJANI BACHIR

Full-Stack Developer

📍 Blida, Algeria

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🌐 GitHub: <https://github.com/Tidjani1Bachir>

🌐 Portfolio: tidjani-bachir.vercel.app



SUMMARY

Full-Stack Developer specializing in building AI-powered web applications, backend systems, and REST APIs using TypeScript, Next.js, and Node.js. Experienced in designing LLM-based agents, database-driven architectures with Prisma and PostgreSQL, and integrating third-party services such as email automation and AI APIs. Open-source contributor with merged PRs in freeCodeCamp.

OPEN SOURCE CONTRIBUTIONS

freeCodeCamp | github.com/freeCodeCamp/freeCodeCamp

- Fixed privacy save button state bug by implementing lodash.isEqual for derived state comparison, per code review by Shaun Hamilton (freeCodeCamp Maintainer, 5+ yrs exp).
- Delivered production-ready code with full test coverage (9/9 tests), successfully merged into main branch.
- PR (Merged) : <https://github.com/freeCodeCamp/freeCodeCamp/pull/67230>

Mongoose ODM | github.com/Automattic/mongoose

- Independently identified missing TypeScript type exports — diagnosis confirmed correct when PR was closed as a duplicate of a simultaneously filed fix by another contributor.
- Submitted PR #16239 with comprehensive type assertion tests across 39 test files (958 assertions); fix validated by maintainers as correct.
- PR: <https://github.com/Automattic/mongoose/pull/16239>

PROJECTS

FlowCRM — AI Support Chatbot:

- **Live App:** <https://flowcrm-ai-support.vercel.app>
- **GitHub :** <https://github.com/Tidjani1Bachir/flowcrm-ai-support>
- Built an AI-powered customer support chatbot using Groq Llama 3.1 with structured JSON response generation, intent detection, and conversation management.
- Designed and implemented a PostgreSQL database schema with Prisma ORM, modeling chat sessions, messages, and lead records with relational integrity.
- Developed an agentic lead-capture workflow that extracts structured customer information (name, email, company) from natural-language conversations.
- Automated lead notification pipelines using Resend, delivering formatted email alerts while preventing duplicate submissions.
- Implemented robust backend validation with Zod, session-based conversation tracking, and API rate limiting to improve reliability and security.

- Built comprehensive automated tests with Vitest and end-to-end validation, covering AI response processing, lead extraction, and API workflows.
- Deployed the application on Vercel with production-ready error handling, database persistence, and scalable API architecture.

Full-Stack E-Commerce Platform — MERN Stack:

- **Live App:** <https://shops-frontend.vercel.app>
- **GitHub (Dockerized) :** <https://github.com/Tidjani1Bachir/fullstack-shopsphere>
- Built a secure full-stack e-commerce platform with JWT authentication, HttpOnly cookies, CSRF protection, and NoSQL injection prevention
- Developed an API-driven React frontend using Redux Toolkit and RTK Query with real-time cache synchronization
- Designed a decoupled system with scalable MongoDB schemas and RTK Query performance optimization
- Integrated PayPal payments using secure SDK-based iframe isolation.
- Built a hybrid favorites system that stored guest data in localStorage and synced seamlessly with MongoDB after login, preserving favorites across devices without disrupting the user experience.
- Dockerized full stack (Node/Express, Vite/Nginx, MongoDB) with Docker Compose and health checks.

Business analysis: <https://tidjani1bachir.github.io/business-analysis>

Product Manager:

- **Live App:** <https://product-manager-chi-eosin.vercel.app>
- **Desktop (.exe):** https://github.com/Tidjani1Bachir/product_manager/releases/download/v2.3.1/product-manager_0.1.0_x64-setup.exe
- Built a React + TypeScript frontend with a REST API-driven Node.js/Express backend and clean frontend/backend separation using Zustand for state management
- Implemented event-driven UI updates with real-time dashboard synchronization without page reloads
- Built a 30-day soft-delete recovery system coordinating frontend state, REST APIs, and SQLite persistence in a consistent workflow
- Integrated Cloudinary image uploads with async API handling and backup local image support
- Built automated PDF generation via server-side pdfmake, generation API with direct file download support
- Containerized the full stack using Docker and Docker Compose, orchestrating a multi-container architecture with an nginx reverse proxy.

Business analysis: <https://tidjani1bachir.github.io/business-analysis>

Product Manager — AWS Edition (DynamoDB + LocalStack):

- **GitHub:** <https://github.com/Tidjani1Bachir/product-manager-dynamodb-localstack-setup>
- Migrated from SQLite (Turso) to AWS DynamoDB by redesigning a scalable multi-entity schema, and built a LocalStack + Docker for seamless deployment switching to AWS deployments with zero code changes.

EXPERIENCE

IT Assistant / Cyber Café Operator Blida, Algeria | ~2 Months

- Provided technical support including software installation and troubleshooting
- Assisted customers with internet services, document processing

EDUCATION

Master of Science in Software Engineering (In Progress)	2024 – Present
<i>University of Saad Dahleb – Blida, Algeria</i>	
Bachelor of Science in Computer Systems	2020 – 2024
<i>University of Saad Dahleb – Blida, Algeria</i>	
• Final Year Project: Lawyer-client management platform using Firebase (backend) and Android Studio / Java	

TECHNICAL SKILLS

<u>Frontend:</u> React, Next.js, TypeScript, Tailwind CSS, Redux Toolkit, RTK Query
<u>Backend:</u> Node.js, Express.js, REST APIs, JWT Auth, Secure File Uploads
<u>Database:</u> MongoDB, Turso (SQLite) , AWS DynamoDB (SDK v3), SQL Server (academic), Oracle PL/SQL (academic) , PostgreSQL , Prisma
<u>AI & APIs</u> :Groq API, Prompt Engineering, AI Agent Development
<u>Testing:</u> Vitest, Playwright, React Testing Library, Postman , Zod
<u>Deployment & Tools:</u> Git/GitHub, Vercel, Render, Cloudinary , Docker, LocalStack (AWS Emulator)

LANGUAGES

Arabic (Native), English (Professional Working), French (Professional Working)
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